

# Endometrial Cancer

## A Review

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**IMPORTANCE** Endometrial cancer accounts for 95% of uterine cancers. Uterine cancers are the fourth most common cancer in women in the US, with an estimated 68 270 new cases and 14 450 cancer deaths projected to occur in 2026. The US mortality rate from uterine cancer rose by 1.6% annually from 2014 to 2023, increasing from 4.8 deaths per 100 000 persons to 5.6 deaths per 100 000 persons.

**OBSERVATIONS** Endometrial cancer develops most commonly due to the effect of estrogen on the uterine lining. Circulating estrogen is produced by adipose cells; rising obesity rates are correlated with an increase in US endometrial cancer incidence from 2013 to 2022 (26.4 per 100 000 in 2013 and 28.8 per 100 000 in 2022). Although 80% of endometrial cancers arise in postmenopausal women, the incidence increased from 2000 to 2019 by 3.0% annually in women aged 20 to 29 years and 3.3% annually in women aged 30 to 39 years. The age-adjusted incidence of endometrial cancer is similar among Black and White women, but mortality was 18.4 per 100 000 among Black women and 8.1 per 100 000 among White women from 2019 to 2023. Many patients (75%-90%) with endometrial cancer present with vaginal bleeding, which should be evaluated with office endometrial biopsy or surgical dilation and curettage for pathologic diagnosis. Initial endometrial cancer management is surgery, including hysterectomy, bilateral salpingo-oophorectomy, and lymph node assessment. Ovarian retention can be considered in select younger patients. Surgical pathology defines histology and molecular subtype, which guide staging and adjuvant therapy. Early-stage, low-risk disease is managed with observation after surgery. Patients with early-stage, high-intermediate risk disease typically undergo adjuvant radiation therapy and patients with aggressive histologic subtypes or advanced-stage disease are treated with adjuvant chemotherapy. First-line chemotherapy for advanced-stage endometrial cancer is carboplatin and paclitaxel plus consideration of anti-programmed cell death protein 1 immunotherapy. Radiation therapy is used for localized recurrences in the pelvis; distant or multifocal recurrences are treated with systemic therapy, including chemotherapy, without or with immunotherapy, or hormonal therapy such as progestins or antiestrogens. Five-year survival is greater than 86% for stage I endometrial cancer, greater than 75% for stage II, greater than 41% for stage III, and greater than 18% for stage IV disease.

**CONCLUSIONS AND RELEVANCE** Endometrial cancer accounts for 95% of uterine cancers, which are the fourth most common cancer in women in the US. Surgery is first-line treatment. Adjuvant radiation is recommended for early-stage, high-intermediate risk endometrial cancer. Adjuvant chemotherapy with carboplatin and paclitaxel is recommended for early-stage disease with aggressive histologic subtypes and advanced-stage disease, which is also treated with immunotherapy in some cases.

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**U**terine cancer, which includes malignancies of the endometrium (95%) and myometrium (5%), is the fourth most common cancer in women in the US, with an estimated 68 270 new cases and 14 450 cancer deaths projected to occur in 2026.<sup>1</sup> Overall, 80% of endometrial cancers in the US occur in postmenopausal women, but incidence increased at higher rates between 2000 and 2019 in premenopausal women.<sup>2</sup> Black women had approximately 2-fold increased mortality compared with White women from 2019 to 2023 (18.4 vs 8.1 deaths per 100 000 persons).<sup>1</sup> The mortality difference<sup>1</sup> may be due, in part, to a higher rate of aggressive histologic subtypes among Black women compared with White women (25.9 vs 11.4 per 100 000 person-years).<sup>3</sup>

More than three-quarters of endometrial cancers present with vaginal bleeding caused by abnormal uterine bleeding, defined as bleeding that is abnormal in regularity, volume, frequency, or duration. Diagnosis is made by examination of endometrial tissue obtained with either office endometrial biopsy or surgical dilation and curettage. First-line treatment is surgery.<sup>4,5</sup> Surgical pathology defines histologic subtype. The most common histologic subtypes of endometrial cancer are endometrioid (87.2%), serous (5.6%), carcinosarcoma (3.9%), clear cell (1.7%), and mixed type (1.7%),<sup>6</sup> with grade 1 and 2 endometrioid pathology classified as nonaggressive subtypes and all others classified as aggressive subtypes. Additionally, the tumor is categorized into 1 of the following 4 molecular subtypes: p53 wild type, mismatch repair deficient (MMR-D), p53 abnormal, or polymerase epsilon (*POLE*) mutated.<sup>7</sup> Pathologic assessment of disease spread combined with histologic subtype and molecular classification defines stage and guides treatment recommendations.<sup>8</sup>

This Review summarizes the epidemiology, clinical presentation, diagnosis, surgical management, histopathologic and molecular characteristics, and adjuvant treatment for endometrial cancer. Given the differences in the epidemiology of endometrial cancer between the US and globally, this article focuses on endometrial cancer in the US. Some common clinical questions and answers are shown in the Box.

## Methods

A PubMed search was performed for English-language articles describing randomized clinical trials, meta-analyses, and systematic reviews of endometrial cancer published between January 1, 2010, and December 15, 2025. The authors selected articles for inclusion, prioritizing recent randomized clinical trials of higher quality based on rigor of study design, adequate sample size, and long-term follow-up. Literature from review of citations in retrieved articles and foundational publications in the field before 2010 were also assessed. Of 2078 articles identified, 107 were included, composed of 31 clinical trials, 19 reviews (11 systematic, 1 umbrella, 7 narrative), 3 meta-analyses, 17 guideline recommendations, and 37 observational and cohort studies. Whenever available, absolute risks are reported to describe outcomes from the included studies.

## Discussion/Observations

### Epidemiology and Risk Factors

Approximately 1 in 32 women in the US develop uterine cancer in their lifetime.<sup>1</sup> Endometrial cancer incidence has increased in the US

### Box. Commonly Asked Questions About Endometrial Cancer

#### What are risk factors and protective factors for endometrial cancer?

The leading modifiable risk factor for nonaggressive histologic subtypes (grade 1 and 2 endometrioid) of endometrial cancer is obesity, which is associated with excess estrogen production by adipose cells. Additional risk factors include nulliparity, tamoxifen therapy, and diabetes. In contrast, oral contraceptives and progesterone therapy, including the levonorgestrel intrauterine device, are associated with decreased risk of endometrial cancer. A total of 5% to 10% of endometrial cancers are hereditary, due to syndromes such as Lynch, Cowden, and Peutz-Jeghers.

#### How does endometrial cancer typically present?

Approximately 70% to 90% of patients with endometrial cancer have vaginal bleeding due to abnormal uterine bleeding, defined as bleeding that is abnormal in regularity, volume, frequency, or duration. Uterine enlargement, pelvic/abdominal pain, or anemia may occur in some women, especially among Black individuals. Endometrial cancer may also be diagnosed following detection of atypical glandular cells on a Papanicolaou test, as an incidental finding of a thickened endometrial lining on imaging, or on pathologic assessment of the uterus after hysterectomy performed for other indications.

#### How is endometrial cancer diagnosed?

Endometrial cancer is diagnosed by sampling the lining of the endometrium, with an endometrial biopsy using a disposable suction curette performed in an outpatient clinic or with a dilation and curettage performed in the operating room.

from 2013 to 2022 (26.4 per 100 000 persons in 2013 and 28.8 per 100 000 persons in 2022).<sup>1</sup> The mortality rate from uterine cancer also increased from 2014 to 2023 (increasing from 4.8 deaths per 100 000 persons to 5.6 deaths per 100 000 persons).<sup>1</sup> The risk of endometrial cancer increases with age; more than 80% of cases occur in postmenopausal women with a median age at diagnosis of 60 years.<sup>1</sup> Increasing incidence is most pronounced in younger women in the US, with an annual rate of increase of 3.3% in women aged 20 to 29 years and 3.0% in women aged 30 to 39 years from 2000 to 2019.<sup>2</sup>

### Hormonal Risk Factors

Nonaggressive (grade 1 and 2 endometrioid) endometrial cancers comprise approximately two-thirds of cases and arise due to unopposed estrogen. Endometrial hyperplasia is a precursor lesion; 1 in 3 women with atypical hyperplasia develop endometrial cancer within 20 years.<sup>9</sup> In a prospective cohort study of 289 women with complex atypical hyperplasia prior to hysterectomy, 42.6% had concurrent endometrial cancer on pathologic review of the hysterectomy specimen.<sup>10</sup>

Obesity is the strongest risk factor for the nonaggressive histologic subtypes of endometrial cancer.<sup>11</sup> Estrogen production by adipose cells is proposed to account for the association between obesity and endometrial cancer.<sup>12</sup> In premenopausal women, the obesity-related increased relative risk per every 5 units of body mass index (BMI; calculated as weight in kilograms divided by height in meters squared) is 1.49 (95% CI, 1.39-1.61).<sup>13</sup> In the Million Women Study, the relative risk was 2.73 in women with a BMI greater than 30 compared with women with a BMI of 22.5 to 24.9.<sup>14</sup> Decreasing BMI in

patients with obesity may lower risk of endometrial cancer. A systematic review of 5 case-control studies reported that 462 of 113 032 patients (0.4%) who underwent bariatric surgery developed endometrial cancer vs 11 997 of 848 864 controls (1.4%) (odds ratio, 0.32 [95% CI, 0.16-0.63]).<sup>15</sup>

Tamoxifen, used for breast cancer risk reduction, has estrogenic effects on the endometrium and is associated with a 2- to 3-fold higher risk of developing endometrial cancer.<sup>16</sup> Endometriosis is also associated with an increased risk of endometrial cancer, with a reported relative risk of 1.66 (95% CI, 1.15-2.41).<sup>17</sup> Nulliparity is associated with increased risk of endometrial cancer; the relative risk of endometrial cancer for parous women was reported in an umbrella review of 171 meta-analyses as 0.66 (95% CI, 0.60-0.74).<sup>13</sup>

In contrast, progesterone therapy is protective for nonaggressive endometrial cancer subtypes and the use of combined estrogen and progesterone-containing oral contraceptives has been associated with a decreased risk of endometrial cancer. An individual participant meta-analysis of 36 observational studies including 27 276 women with endometrial cancer and 115 743 control women reported that 10 years of oral contraceptive use was associated with a reduction in the absolute risk of endometrial cancer before age 75 years from 2.3 to 1.3 cases per 100 women.<sup>18</sup> Progesterone therapy in combination with estrogen in postmenopausal hormone replacement therapy is also associated with a decreased risk of endometrial cancer.<sup>19,20</sup> In a pooled analysis of individual-level data from 4 cohort and 14 case-control studies with 8801 endometrial cancer cases and 15 357 controls, progesterone therapy with an intrauterine device (IUD) was inversely related to endometrial cancer risk, with a pooled odds ratio of 0.81 (95% CI, 0.74-0.90).<sup>21</sup>

Diabetes is an independent risk factor for endometrial cancer, with a global systematic review and meta-analysis of 29 cohort studies (17 prospective and 12 retrospective) reporting a relative risk of 1.89 (95% CI, 1.46-2.45).<sup>22</sup> Two retrospective cohort studies reported that glucagon-like peptide-1 receptor agonist use was associated with a decreased risk of endometrial cancer,<sup>23,24</sup> but a systematic review and meta-analysis did not report an association.<sup>25</sup>

### Genetic Risk Factors

Multiple hereditary cancer syndromes increase the risk of developing endometrial cancer. Women with Lynch syndrome, caused by an inherited pathogenic variant in one of the multiple MMR proteins, have a 30% to 50% lifetime risk of developing endometrial cancer.<sup>26</sup> Risk of developing endometrial cancer is 5% to 22% for individuals with Cowden syndrome (germline *PTEN* pathogenic variant) and 9% for those with Peutz-Jeghers syndrome (germline *STK11* pathogenic variant).<sup>26,27</sup> Current National Comprehensive Cancer Network (NCCN) guidelines advise that annual endometrial cancer screening with endometrial biopsy be considered for individuals with Lynch, Cowden, or Peutz-Jeghers syndrome.<sup>26,27</sup> Per NCCN, based on limited data demonstrating a slightly increased risk of endometrial cancer among individuals with *BRCA* inherited pathogenic variants,<sup>28</sup> hysterectomy may be considered at the time of ovarian cancer risk-reducing surgery to remove the ovaries and fallopian tubes in women with a *BRCA* pathogenic variant.<sup>27</sup>

### Clinical Presentation and Diagnostic Evaluation

Endometrial cancer presents as abnormal vaginal bleeding in 75% to 90% of patients, including 90% of postmenopausal women.<sup>29</sup>

In a retrospective study of 404 Black women diagnosed with endometrial cancer from 2014 to 2020, additional symptoms included fibroids/uterine enlargement (70%), pelvic/abdominal pain (38%), and anemia (30%).<sup>30</sup> Rarely, endometrial cancer may be detected on cervical cytology Papanicolaou tests. A retrospective review of 247 131 cervical cytology Papanicolaou tests at a single academic center included 1021 (0.41%) with atypical glandular cells, of which 5.1% (34/662) were diagnosed with endometrial cancer.<sup>31</sup> Endometrial cancer can also be incidentally identified on abdominal imaging such as computed tomography, magnetic resonance imaging (MRI), or ultrasonography performed for another indication. A retrospective study reported that 3.2% of women (3/94) with increased endometrial thickness that was incidentally detected on imaging had endometrial cancer.<sup>32</sup> Additionally, endometrial cancer is diagnosed based on pathologic evaluation of the uterus in less than 1% of hysterectomies performed for presumed benign indications.<sup>33</sup>

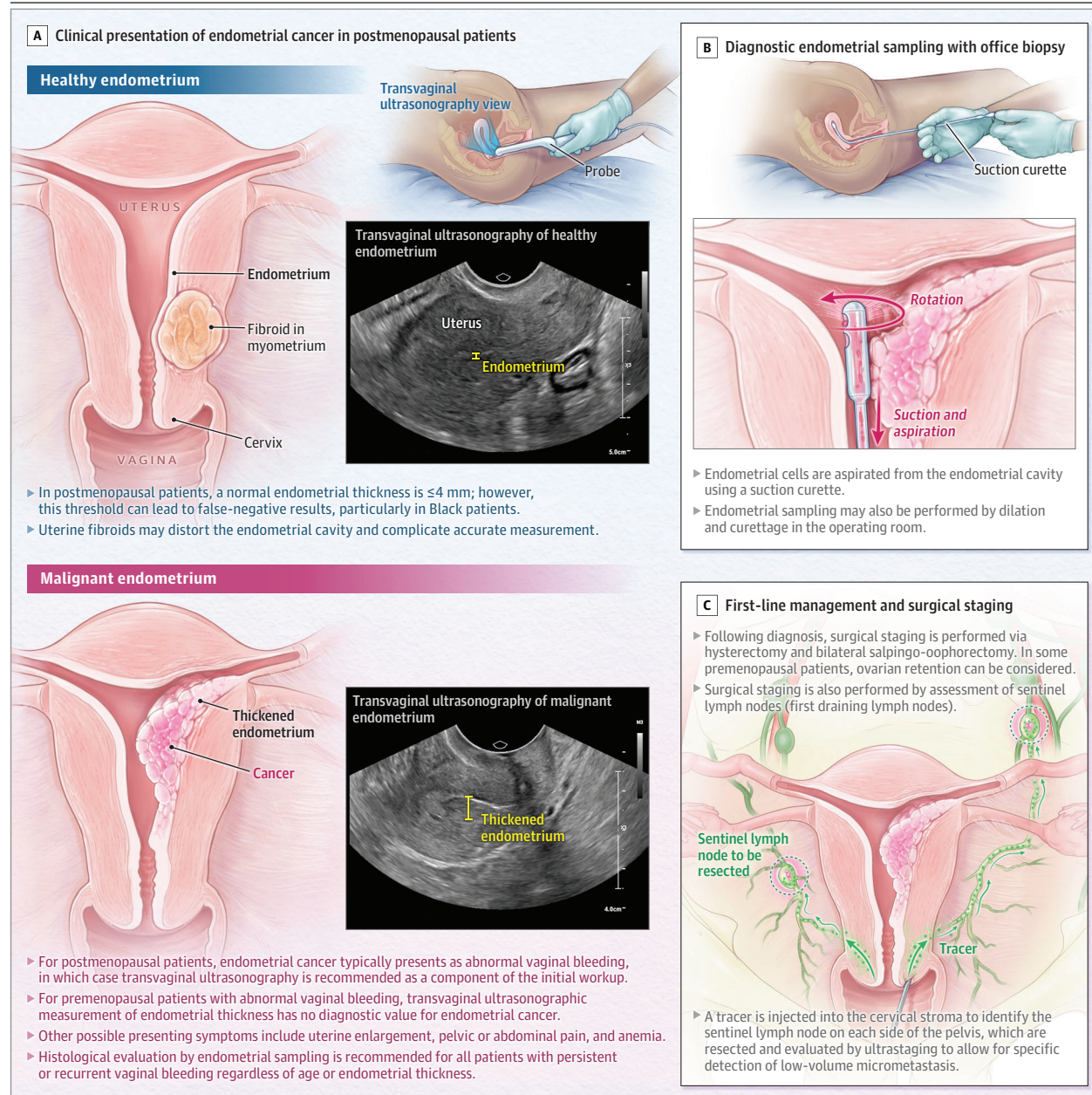
The American College of Obstetricians and Gynecologists (ACOG) committee expert opinion recommends transvaginal ultrasonography as the initial approach for postmenopausal women with a single episode of vaginal bleeding (Figure 1); in contrast, transvaginal ultrasonography has no diagnostic value for vaginal bleeding in premenopausal women and should not be performed, based on the ACOG committee opinion.<sup>29</sup> In postmenopausal women, an endometrial thickness of 4 mm or less has a greater than 99% negative predictive value for endometrial cancer based on a pooled analysis of 6790 women in 4 observational studies.<sup>29</sup> However, a retrospective study of 1494 Black women who underwent pelvic ultrasonography prior to hysterectomy reported a 9.5% false-negative result rate for detecting endometrial cancer using the 4-mm threshold endometrial thickness on pelvic ultrasonography.<sup>34</sup> Reasons for this racial difference may include the increased prevalence of uterine fibroids among Black women that may make accurate endometrial measurements more difficult and the greater frequency of aggressive histologic subtypes among Black women.<sup>30,34,35</sup> Guidelines do not currently recommend using a different endometrial thickness threshold in Black women. Moreover, data indicate that in all women with persistent vaginal bleeding, the aggressive subtypes of endometrial cancer cannot be excluded based on an endometrial thickness less than 4 mm.<sup>36</sup> Therefore, ACOG recommends histologic evaluation for persistent or recurrent bleeding regardless of endometrial thickness.<sup>29</sup>

A meta-analysis of 9 cohort studies that included 4751 asymptomatic women who underwent transvaginal ultrasonography reported a mean prevalence of endometrial hyperplasia/cancer of 1.7% with endometrial thickness less than 11 mm vs 6.5% with endometrial thickness greater than or equal to 11 mm (relative risk for endometrial cancer, 2.59 [95% CI, 1.66-4.05]).<sup>37</sup> Based on ACOG recommendations, incidental identification of endometrial thickness greater than 4 mm in a postmenopausal woman without vaginal bleeding need not be further evaluated with biopsy or serial ultrasonography, although individualized assessment based on clinical presentation and risk factors is recommended.<sup>29</sup>

Women being evaluated for endometrial cancer should undergo diagnostic tissue sampling of the endometrial lining (Figure 1), with endometrial biopsy using a disposable suction curette<sup>38</sup> performed in an outpatient setting or cervical dilation and endometrial curettage (dilation and curettage) performed in an operating



Figure 1. Initial Imaging, Diagnosis, and Surgical Staging of Endometrial Cancer



A, Schematic and ultrasonography appearance of normal and malignant endometrium. B, Biopsy of the endometrial lining for diagnosis can either be performed with a catheter suction curette in the office (shown) or via dilation and curettage in the operating room. C, First-line management following

diagnosis is surgical staging, including hysterectomy, bilateral salpingo-oophorectomy, and lymph node assessment. Lymph node assessment is most commonly performed by the sentinel lymph node method (shown).

room. A systematic review and meta-analysis of 12 cross-sectional studies including 1607 women with clinical suspicion for endometrial cancer reported that compared with the reference standard of histopathology of hysterectomy specimens, office-based endometrial biopsy had a sensitivity of 0.774 (95% CI, 0.565-0.900) and specificity of 0.985 (95% CI, 0.927-0.997) and dilation and curettage had a sensitivity of 0.880 (95% CI, 0.281-0.993) and specificity of 0.984 (95% CI, 0.956-0.995) for diagnosis of endometrial cancer.<sup>39</sup> If endometrial biopsy is inconclusive, dilation and curettage for additional tissue should be performed.

### Surgical Management

Surgery is first-line treatment for endometrial cancer if disease is resectable and comorbidities do not preclude surgical treatment.<sup>8</sup> To assess feasibility of resection, per NCCN guidelines, chest radiographic imaging to assess for lung metastasis can be considered in all patients and computed tomographic imaging of the chest, abdomen, and pelvis is recommended for patients with aggressive histologic subtypes.<sup>8,40</sup> Surgery can be performed via laparotomy or minimally invasively with a conventional laparoscopy or a robot-assisted approach. NCCN guidelines recommend a minimally invasive surgical

procedure due to lower frequency of surgical complications such as postoperative ileus or cardiac arrhythmia, decreased length of stay following surgery, and lower cost of care compared with laparotomy, with no adverse effect on oncologic outcomes.<sup>4,8,41,42</sup>

Surgical treatment involves a total hysterectomy (uterus plus cervix), bilateral salpingo-oophorectomy, and lymph node assessment. However, among premenopausal women with clinical stage I (limited to the uterus) histologic grade 1 endometrial cancer, NCCN guidelines indicate that retaining the ovaries can be considered. An analysis of 1998 to 2012 National Cancer Database data from 15 648 women younger than 50 years with stage I endometrioid endometrial adenocarcinoma reported that ovarian preservation was not independently associated with survival (hazard ratio [HR] for mortality, 0.94 [95% CI, 0.65-1.37]).<sup>43</sup>

Lymph node assessment is performed via sentinel lymph node dissection, in which a tracer is injected into the cervix and visualized intraoperatively to identify the first draining (sentinel) lymph node on each side of the pelvis. Resected sentinel lymph nodes are evaluated by ultrastaging (thin serial sections for pathology review) for detection of low-volume micrometastasis.<sup>44</sup> Sentinel lymph node dissection is associated with a higher rate of detection of lymph node metastases and decreased morbidity compared with full pelvic lymphadenectomy, in which nodal tissue is removed from the external iliac, internal iliac and obturator nodal basins.<sup>5,45,46</sup> In a prospective cohort study of 385 patients with clinical stage I endometrial cancer of all grades and histologic subtypes,<sup>5</sup> patients underwent sentinel lymph node mapping followed by a full pelvic lymphadenectomy. Sentinel lymph node identification occurred in 86% of patients; sentinel lymph node dissection had a sensitivity of 97.2% (95% CI, 85%-100%) and a negative predictive value of 99.6% (95% CI, 97.9%-100%) for detecting lymph node metastases.<sup>5</sup> If sentinel lymph node mapping is not successful, NCCN guidelines recommend full pelvic lymphadenectomy.<sup>8</sup>

## Staging

Surgical pathology results define the stage of endometrial cancer. The 2023 International Federation of Gynecology and Obstetrics (FIGO) staging system for endometrial cancer<sup>47</sup> is a 4-stage system based on TNM (tumor, nodes, metastasis) criteria plus histologic classification and molecular subtyping (Table 1).<sup>7</sup> However, current NCCN guidelines are based on the 2009 FIGO staging system<sup>48</sup> rather than the 2023 FIGO staging system because it has not been fully adopted.<sup>8</sup>

Molecular classification of endometrial cancers provides prognostic information and therefore is recommended by NCCN guidelines for all patients diagnosed with endometrial cancer. The Cancer Genome Atlas classified 373 endometrial cancer cases into 4 molecular subtypes based on tumor genetic sequencing and prognosis (Table 2).<sup>49</sup> These molecular subtypes have been further refined in the Proactive Molecular Risk Classifier for Endometrial Cancer (ProMisE) study (Table 2), which uses immunohistochemistry for p53 and MMR protein expression combined with directed sequencing of the DNA *POLE* gene to classify endometrial cancer into the subtypes of p53 wild type, MMR-D, p53 abnormal, and *POLE* mutated.<sup>7</sup> MMR immunohistochemistry also helps identify patients who may have germline pathogenic variants indicative of Lynch syndrome, and therefore should be performed on tumor tissue for all endometrial cancers.<sup>26</sup> However, *POLE* classification requires gene sequencing of the gene regions that frequently have pathogenic

**Table 1. International Federation of Gynecology and Obstetrics 2023 Staging of Endometrial Cancer<sup>a</sup>**

| Stage                               | Description                                                                                                                                                                                                                         |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Stage I                             | Confined to the uterine corpus and ovary                                                                                                                                                                                            |
| IA                                  | Disease limited to the endometrium or nonaggressive histological type, ie, low-grade endometrioid, with invasion of less than half of myometrium with no or focal lymphovascular space involvement (LVSI) or good prognosis disease |
| IA1                                 | Nonaggressive histological type limited to an endometrial polyp or confined to the endometrium                                                                                                                                      |
| IA2                                 | Nonaggressive histological types involving less than half of the myometrium with no or focal LVSI                                                                                                                                   |
| IA3                                 | Low-grade endometrioid carcinomas limited to the uterus and ovary                                                                                                                                                                   |
| IAm <sup>POLEmut</sup> <sup>b</sup> | POLE-mutated endometrial carcinoma, confined to the uterine corpus or with cervical extension, regardless of the degree of LVSI or histological type                                                                                |
| IB                                  | Nonaggressive histological types with invasion of half or more of the myometrium with no or focal LVSI                                                                                                                              |
| IC                                  | Aggressive histological types limited to a polyp or confined to the endometrium                                                                                                                                                     |
| Stage II                            | Invasion of cervical stroma without extrauterine extension or with substantial LVSI or aggressive histological types with myometrial invasion                                                                                       |
| IIA                                 | Invasion of the cervical stroma of nonaggressive histological types                                                                                                                                                                 |
| IIB                                 | Substantial LVSI of nonaggressive histological types                                                                                                                                                                                |
| IIC                                 | Aggressive histological types with any myometrial involvement                                                                                                                                                                       |
| IICm <sup>p53abn</sup> <sup>c</sup> | p53 abnormal endometrial carcinoma confined to the uterine corpus with any myometrial invasion, with or without cervical invasion, and regardless of the degree of LVSI or histological type                                        |
| Stage III                           | Local and/or regional spread of the tumor of any histological subtype                                                                                                                                                               |
| IIIA                                | Invasion of uterine serosa, adnexa, or both by direct extension or metastasis                                                                                                                                                       |
| IIIA1                               | Spread to ovary or fallopian tube (except when meeting stage IA3 criteria)                                                                                                                                                          |
| IIIA2                               | Involvement of uterine subserosa or spread through the uterine serosa                                                                                                                                                               |
| IIIB                                | Metastasis or direct spread to the vagina and/or to the parametria or pelvic peritoneum <sup>d</sup>                                                                                                                                |
| IIIB1                               | Metastasis or direct spread to the vagina and/or the parametria                                                                                                                                                                     |
| IIIB2                               | Metastasis to the pelvic peritoneum                                                                                                                                                                                                 |
| IIIC                                | Metastasis to the pelvic or para-aortic lymph nodes or both                                                                                                                                                                         |
| IIIC1                               | Metastasis to the pelvic lymph nodes (IIIC1i = micrometastasis; IIIC1ii = macrometastasis)                                                                                                                                          |
| IIIC2                               | Metastasis to para-aortic lymph nodes up to the renal vessels, with or without metastasis to the pelvic lymph nodes (IIIC1i = micrometastasis; IIIC1ii = macrometastasis)                                                           |
| Stage IV                            | Spread to the bladder mucosa and/or intestinal mucosa and/or distant metastasis                                                                                                                                                     |
| IVA                                 | Invasion of the bladder mucosa and/or the intestinal/bowel mucosa                                                                                                                                                                   |
| IVB                                 | Abdominal peritoneal metastasis beyond the pelvis                                                                                                                                                                                   |
| IVC                                 | Distant metastasis, including metastasis to any extra- or intra-abdominal lymph nodes above the kidney vessels, lungs, liver, brain, or bone                                                                                        |

<sup>a</sup> Originally published in the *International Journal of Gynecology and Obstetrics*.<sup>47</sup> Modified to include molecular stages as a single table.

<sup>b</sup> Molecular finding of polymerase epsilon (*POLE*) mutation for surgical stage I and II endometrial cancer down-stages to IAm due to the favorable prognostic outcomes with this molecular alteration.

<sup>c</sup> Molecular finding of p53 abnormal for surgical stage I and II endometrial cancer up-stages to IICm due to the poor prognostic outcomes with this molecular alteration.

<sup>d</sup> Parametria are the fat and connective tissue lateral to the uterus on both sides.

Table 2. Molecular Subtypes of Endometrial Cancer

| ProMisE Classification (estimated frequency) <sup>7</sup>                  | p53 wild type (50.4%)                                                                                                                                | MMR-deficient (MMR-D) (28.1%)                                                                                          | p53 abnormal (12.2%)                                                                                                                    | POLE mutated (9.3%)                                                     |
|----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| The Cancer Genome Atlas Classification (estimated frequency) <sup>49</sup> | Copy number low or no specific molecular profile (39%-46%)                                                                                           | Microsatellite instability hypermutated (28%-29%)                                                                      | Copy number high (26%-27%)                                                                                                              | POLE ultramutated (7%-10%)                                              |
| Histologic subtypes                                                        | Low-grade endometrioid                                                                                                                               | Low-grade endometrioid, high-grade endometrioid, and undifferentiated/dedifferentiated                                 | Serous, high-grade endometrioid, clear cell, carcinosarcoma, and mixed                                                                  | High-grade endometrioid, serous, and mixed                              |
| Prognosis (estimated 5-y survival) <sup>50</sup>                           | Intermediate (65%-80%)                                                                                                                               | Intermediate (75%-80%)                                                                                                 | Lowest (35%-50%)                                                                                                                        | Highest (90%-93%)                                                       |
| Treatment implications <sup>a</sup>                                        | Consider stage in determining treatment: radiation for early-stage disease and chemotherapy with or without immunotherapy for advanced-stage disease | Enhanced sensitivity to immunotherapy: consider immunotherapy in addition to or instead of stage-based recommendations | Most effective treatment is cytotoxic chemotherapy even in early-stage disease with or without immunotherapy for advanced-stage disease | Excellent prognosis may allow deescalation of standard adjuvant therapy |

Abbreviations: POLE, polymerase epsilon; ProMisE, Proactive Molecular Risk Classifier for Endometrial Cancer.

<sup>a</sup> See Table 3 for details of current chemotherapy and immunotherapy agents.

variants and therefore is not always performed due to lack of pathology laboratory resources.<sup>51</sup>

### Management of Localized Endometrial Cancer for Individuals Desiring Conception

The increased endometrial cancer incidence among patients younger than 40 years<sup>2,52</sup> has led to a focus on uterine-sparing options for early-stage, nonmyoinvasive (confined to the endometrium) disease, primarily using high-dose progesterone therapy, including progestins such as megestrol acetate or a levonorgestrel-releasing IUD. Multidisciplinary assessment is recommended when fertility-preserving management of endometrial cancer is being considered, including expert pathology review and MRI to ensure nonmyoinvasive disease. MRI has 87% sensitivity and a 92% positive predictive value for myometrial invasion.<sup>53</sup> Clinicians should discuss fertility considerations in cancer treatment planning for patients younger than 40 years.<sup>54,55</sup> Additionally, NCCN guidelines recommend evaluation with a reproductive endocrinologist.<sup>8</sup> For patients who meet criteria for fertility-preserving treatment, progesterone therapy is administered followed by endometrial sampling every 3 months with the goal of inducing endometrial cancer regression. A prospective phase 2 trial of 57 patients with a median age of 48 years using a levonorgestrel-releasing IUD for uterine-sparing management of endometrial hyperplasia (36 patients) and grade 1 endometrioid endometrial cancer (21 patients) reported a 66.7% rate of cancer regression at 12 months.<sup>56</sup> This study is most reflective of the current patient population and treatment approaches, but retrospective studies and meta-analyses have also demonstrated the efficacy and safety of progesterone treatment delivered orally rather than by IUD in this setting.<sup>57-59</sup> Hysterectomy is recommended if endometrial cancer regression is not achieved after 12 months or when childbearing is completed.<sup>8</sup>

### Management of Localized Endometrial Cancer for Individuals Ineligible for Surgery

Patients with uterus-confined endometrial cancer based on imaging who are ineligible for surgery due to medical comorbidities can be treated with pelvic radiation or progesterone therapy; high-dose-rate brachytherapy with or without external beam radiation is considered first-line treatment.<sup>60</sup> In a systematic review including 25

observational studies and 2694 patients with low-grade, clinical early-stage endometrial cancer treated with radiation therapy alone, 5-year disease-specific survival, which was reported in 49.1% of patients, was 78.5% (95% CI, 74.5%-82.5) and 5-year overall survival, reported in 75.2% of patients, was 53.2% (95% CI, 49.3%-57.1%).<sup>61</sup> A National Cancer Database retrospective cohort study that included 1036 patients with stage I to III (89% had stage I or II) endometrioid endometrial cancer who were unable to undergo surgery due to medical comorbidities reported no difference in overall survival between those treated with progesterone therapy vs radiation; unadjusted 5-year overall survival was 45.4% in the progesterone treatment group and 37% in the radiation therapy group ( $P = .11$ ).<sup>62</sup>

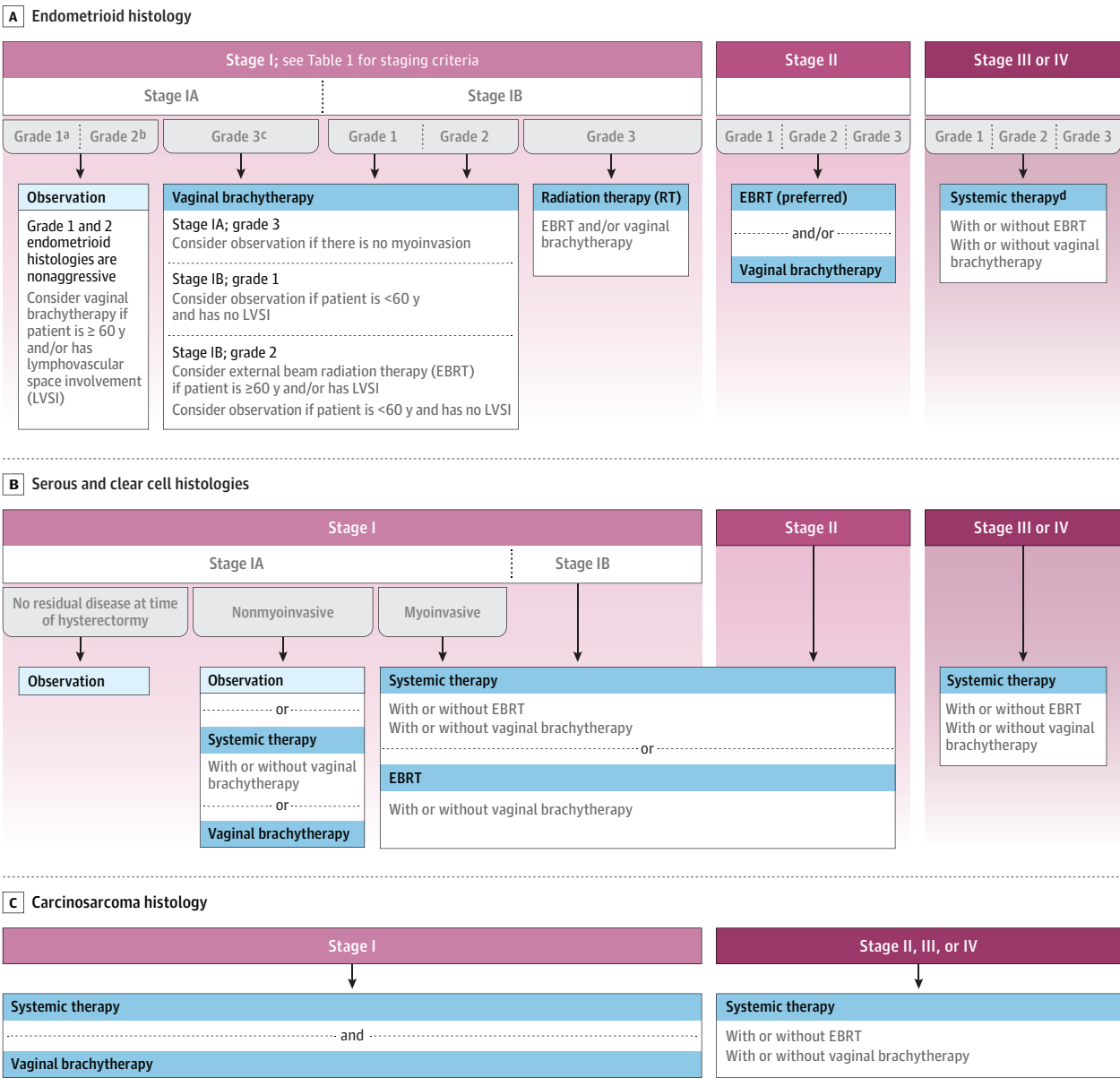
### Adjuvant Therapy for Early-Stage Disease

Following surgery, adjuvant therapy may be administered based on histopathologic and molecular characteristics of the tumor (Figure 2) and risk factors for recurrence, which occurs in approximately 20% of patients with early-stage disease.<sup>63,64</sup> Factors that confer a higher likelihood of recurrence for endometrioid endometrial cancer include older age, lymphovascular space invasion (presence of tumor cells in the lymphatic channels in the uterus), higher tumor grade, and greater depth of myometrial invasion.<sup>63-65</sup> Women who meet criteria for disease classified as high-intermediate risk based on these risk factors benefit from postoperative vaginal brachytherapy (Figure 2).<sup>66</sup> For women with early-stage endometrioid endometrial cancer without high-intermediate risk factors, observation is recommended (Figure 2).<sup>67</sup>

For stage I endometrial cancer, the aggressive histologic subtypes of serous, clear cell, and carcinosarcoma have a higher risk of recurrence and a poorer prognosis than the nonaggressive subtypes.<sup>68,69</sup> Adjuvant chemotherapy with carboplatin and paclitaxel improves both progression-free survival (PFS) and overall survival for patients with serous endometrial cancer.<sup>69</sup> For patients with tumors with the molecular classification of p53 abnormal, adjuvant chemotherapy with carboplatin and paclitaxel resulted in improved recurrence-free survival.<sup>70</sup> Per NCCN guidelines, adjuvant radiation therapy may be considered in addition to systemic chemotherapy for these aggressive histologic subtypes with early-stage disease (Figure 2).<sup>8</sup> It may be possible to deescalate adjuvant



Figure 2. Algorithms to Describe Adjuvant Treatment for Surgically Staged Disease



Adjuvant therapy based on surgical stage per National Comprehensive Cancer Network (NCCN) guidelines is shown. Of note, NCCN guidelines remain based on 2009 International Federation of Gynecology and Obstetrics staging at the time of this publication. Category 2B recommendations have not been included.

<sup>a</sup>Cells are well differentiated.

<sup>b</sup>Cells are moderately differentiated.

<sup>c</sup>Cells are poorly differentiated.

<sup>d</sup>Systemic therapy involves chemotherapy with or without immunotherapy for primary adjuvant therapy (see Table 3).

therapy in a subset of patients with tumors of aggressive histology, such as those without residual disease on hysterectomy specimen or negative peritoneal cytology and no myometrial invasion (Figure 2).<sup>71</sup>

**Therapy for Advanced-Stage or Recurrent Disease**

First-line treatment for stage III or IV endometrial cancer is chemotherapy with carboplatin and paclitaxel,<sup>72-78</sup> given after surgery for resectable disease and either before or without surgery when cancer is initially unresectable. Carboplatin and paclitaxel are also first-line treatment for recurrent endometrial cancer if chemotherapy was not administered in the adjuvant setting. The addition of immunotherapy to standard chemotherapy in these settings yields improved progression-free survival for all patients with advanced or recurrent endometrial cancer, with specific benefit among patients with MMR-D (Table 3).<sup>74-78,81,82,85</sup> Two randomized clinical trials, GY018 and RUBY, used immune checkpoint inhibitors (pembrolizumab or dostarlimab, respectively, both of which bind to programmed cell death protein 1 on T cells) in addition to carboplatin and paclitaxel chemotherapy for patients with stage III or IV or

Table 3. Systemic Therapies for Endometrial Cancer

| Source                                                                                                                                                                                                                             | Regimen                                                                  | Targets                             | Trial type                     | Comparator                                                                                                                                                                      | No. of patients                                          | Primary end point                                | Results                                                                                                                                                                                                                     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|-------------------------------------|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>First-line therapy: advanced stage or recurrent endometrial cancer</b>                                                                                                                                                          |                                                                          |                                     |                                |                                                                                                                                                                                 |                                                          |                                                  |                                                                                                                                                                                                                             |
| Matei et al, <sup>72</sup> 2019; Matei et al, <sup>73</sup> 2025                                                                                                                                                                   | Radiation with sensitizing cisplatin followed by carboplatin, paclitaxel |                                     | Phase III, randomized          | Carboplatin, paclitaxel                                                                                                                                                         | 813                                                      | Relapse-free survival                            | Relapse-free survival: 59% vs 58% (HR, 0.9 [90% CI, 0.74-1.1])<br>OS: Median OS not reached in either group (HR for death, 1.05 [95% CI, 0.82-1.34])                                                                        |
| Eskander et al, <sup>74</sup> 2023; Eskander et al, <sup>75</sup> 2025                                                                                                                                                             | Carboplatin, paclitaxel, pembrolizumab                                   | PD-1                                | Phase III, randomized          | Carboplatin, paclitaxel                                                                                                                                                         | 816                                                      | PFS                                              | MMR-D: 74% vs 38% (HR 0.3 [95% CI, 0.19-0.48])<br>MMR-P: 13.1 vs 8.7 mo (HR, 0.54 [95% CI, 0.41-0.71])<br>OS data immature in both cohorts                                                                                  |
| Mirza et al, <sup>76</sup> 2023; Powell et al, <sup>77</sup> 2024                                                                                                                                                                  | Carboplatin, paclitaxel, dostarlimab                                     | PD-1                                | Phase III, randomized          | Carboplatin, paclitaxel                                                                                                                                                         | 494                                                      | PFS, OS                                          | MMR-D: PFS: 61.4% vs 15.7% (HR, 0.28 [95% CI, 0.16-0.50]); OS: 22.6% vs 53.8% (HR, 0.32 [95% CI, 0.17-0.63])<br>MMR-P: PFS: 28.4% vs 18.8% (HR, 0.76 [95% CI, 0.59-0.98]); OS: 50.5% vs 59.2% (HR, 0.79 [95% CI 0.60-1.04]) |
| Fader et al, <sup>78</sup> 2018                                                                                                                                                                                                    | Carboplatin, paclitaxel, trastuzumab                                     | ERBB2 (formerly HER2/neu)           | Phase II, randomized           | Carboplatin, paclitaxel                                                                                                                                                         | 61                                                       | PFS                                              | PFS: 12.6 mo vs 8 mo (HR, 0.44 [90% CI, 0.26-0.76])                                                                                                                                                                         |
| <b>Subsequent lines of therapy: recurrent endometrial cancer</b>                                                                                                                                                                   |                                                                          |                                     |                                |                                                                                                                                                                                 |                                                          |                                                  |                                                                                                                                                                                                                             |
| Meric-Bernstam et al, <sup>79</sup> 2024                                                                                                                                                                                           | Trastuzumab deruxtecan                                                   | ERBB2 (formerly HER2/neu)           | Phase II, single group, basket | None                                                                                                                                                                            | 40 (endometrial cohort)                                  | Objective response rate                          | 57.5%                                                                                                                                                                                                                       |
| O'Malley et al, <sup>80</sup> 2022                                                                                                                                                                                                 | Pembrolizumab                                                            | PD-1                                | Phase II, single group, basket | None                                                                                                                                                                            | 90                                                       | Objective response rate                          | 48%                                                                                                                                                                                                                         |
| Oaknin et al, <sup>81</sup> 2022                                                                                                                                                                                                   | Dostarlimab                                                              | PD-1                                | Phase I, single group, basket  | None                                                                                                                                                                            | Cohort A1, MMR-D/MSI-H: 129<br>Cohort A2, MMR-P/MSS: 161 | Objective response rate and duration of response | Objective response rate of 43.5% vs 14.1%<br>Median duration of response not reached in either cohort                                                                                                                       |
| Makker et al, <sup>82</sup> 2023                                                                                                                                                                                                   | Pembrolizumab and lenvatinib                                             | PD-1, VEGFR, FGFR, PDGFRα, RET, KIT | Phase III, randomized          | Physician choice of doxorubicin or paclitaxel                                                                                                                                   | 827                                                      | PFS and OS                                       | PFS: 7.3 mo vs 3.8 mo (HR, 0.56 [95% CI, 0.48-0.66])<br>OS: 18.7 mo vs 11.9 mo (HR, 0.65 [95% CI, 0.55-0.77])                                                                                                               |
| Slomovitz et al, <sup>83</sup> 2020                                                                                                                                                                                                | Everolimus and letrozole                                                 | PI3K/AKT/mTOR pathway               | Phase II, randomized           | Medroxyprogesterone acetate and tamoxifen                                                                                                                                       | 74                                                       | Objective response rate                          | 22% vs 25%; objective response rate with everolimus and letrozole, 47%                                                                                                                                                      |
| Mileshkin et al, <sup>84</sup> 2019                                                                                                                                                                                                | Anastrozole                                                              | Estrogen/progesterone               | Phase II, single group, basket | None                                                                                                                                                                            | 82                                                       | Clinical benefit (CR, PR, SD)                    | 44%                                                                                                                                                                                                                         |
| Abbreviations: CR, complete response; FGFR, fibroblast growth factor receptor; HR, hazard ratio; MMR-D, mismatch repair deficient; MMR-P, mismatch repair proficient; OS, overall survival; PD-1, programmed cell death protein 1; |                                                                          |                                     |                                | PDGFRα, platelet-derived growth factor receptor-α; PFS, progression-free survival; PR, partial response; SD, stable disease; VEGF, vascular endothelial growth factor receptor. |                                                          |                                                  |                                                                                                                                                                                                                             |

recurrent endometrial cancer (Table 3).<sup>74-77</sup> In both trials, patients were stratified into 2 prespecified subgroups, MMR-D and MMR proficient (MMR-P). In GYO18,<sup>74,75</sup> in the MMR-D subgroup, median progression-free survival was not reached for the pembrolizumab group vs 7.6 months for the placebo group (HR, 0.30 [95% CI, 0.19-0.48]). The MMR-P cohort also showed benefit in the pembrolizumab group, with median progression-free survival of 13.1 months vs 8.7 months in the placebo group (HR, 0.54 [95% CI, 0.41-0.71]); overall survival data are not yet available. In the RUBY trial,<sup>76,77</sup>

similar findings were noted: in the MMR-D subgroup, the median progression-free survival was 61.4 months with dostarlimab vs 15.7 months in the placebo group (HR, 0.28 [95% CI, 0.16-0.50]). The MMR-P cohort also showed benefit with a progression-free survival of 28.4 months with dostarlimab and 18.8 months in the placebo group (HR, 0.76 [95% CI, 0.59-0.98]).

For patients with stage III or IV endometrial cancer, radiation therapy can be considered in addition to systemic therapy in the first-line setting for patients who are at high risk for localized disease



recurrence, such as patients with more extensive pelvic involvement at time of diagnosis.<sup>72,86</sup> In the subset of stage III endometrial cancer with nonaggressive subtypes, radiation can sometimes be used instead of systemic therapy.<sup>8</sup>

Multiple systemic therapy options are available for recurrent disease, including targeted and immunotherapy-based regimens (Table 3).<sup>79-84</sup> Treatment is selected based on tumor histology, protein expression profile, molecular profile, and tumor-specific pathogenic variants. Potentially targetable proteins include *ERBB2* (formerly *HER2/neu*) receptor<sup>79,87</sup> and estrogen or progesterone receptors.<sup>84</sup> Multiple immunotherapies are approved for MMR-D and tumor mutational burden-high tumors, including pembrolizumab, dostarlimab, avelumab, and nivolumab.<sup>80</sup> Next-generation sequencing of the tumor can also reveal targetable gene alterations, such as *PIK3CA* pathogenic variants.<sup>83</sup>

### Novel Approaches to Treatment

Molecular classification is increasingly being used to refine adjuvant treatment options for patients with endometrial cancer. European Society guidelines incorporate molecular classification into endometrial cancer treatment algorithms,<sup>88</sup> although this is not yet integrated into NCCN guidelines. A trial of 564 patients with newly diagnosed high-intermediate risk endometrioid endometrial cancer reported that molecular-integrated, risk profile-based adjuvant treatment including observation for patients with *POLE*-mutated tumors was noninferior to standard care therapy (vaginal brachytherapy).<sup>89,90</sup> Additional studies are ongoing globally, including a platform of 4 international clinical trials utilizing molecular subtype to tailor adjuvant therapy.<sup>91</sup>

Antibody-drug conjugates are also being investigated for treatment of endometrial cancer.<sup>92</sup> The US Food and Drug Administration (FDA) has approved and granted breakthrough therapy designation for antibody-drug conjugates targeting *ERBB2* and folate receptor alpha, and other antibody-drug conjugates are currently in development.<sup>79,93</sup> Adoptive T-cell therapies, which involve administration of T cells targeting antigens expressed on the tumor cells, are in early stages of development for endometrial cancer and may include modified T-cell receptors, chimeric antigen receptor T cells, or tumor infiltrating lymphocytes.<sup>94-96</sup>

### Prognosis

Based on the Surveillance, Epidemiology, and End Results Summary Stage, the 5-year cancer-specific survival rate for uterine cancer is 81%, with 90% 5-year survival for localized disease (uterus confined), 72% for regional disease (pelvic spread), and 22% for distant disease (outside the pelvis).<sup>97</sup>

Molecular classifiers stratify survival, with those with *POLE*-mutated tumors having the most favorable outcomes (90%-93% 5-year survival), MMR-D and p53 wild type demonstrating intermediate 5-year survival (75%-80% and 65%-80%, respectively), and p53 abnormal having the poorest 5-year survival (35%-50%) (Table 2).<sup>47,50</sup> Prognosis for the 2023 FIGO stages was determined in a retrospective cohort study of 129 146 patients with endometrial cancer in the National Cancer Database.<sup>98</sup> Survival was greater than 88% for stage IA disease and greater than 86% for stage IB and IC disease, greater than 85% for stage IIA or IIB disease and 75.4% for stage IIC disease, 41% to 62% for stage III disease, and 18% to 33% for stage IV disease.<sup>98</sup>

Among patients with endometrial cancer, Black women have a higher incidence-based mortality compared with White women due to factors including different presenting symptoms that result in delayed diagnosis, higher frequency of aggressive histologic subtypes, and disparities in delivery of guideline-concordant care. Among 2607 Medicaid beneficiaries from 2008 to 2019, 11.3% of Black patients experienced delayed diagnosis compared with 8.3% of White patients (adjusted odds ratio [OR], 1.71 [95% CI, 1.27-2.29]) and 10.1% of Black patients did not receive guideline-recommended diagnostic procedures compared with 5.0% of White patients (adjusted OR, 1.94 [95% CI, 1.40-2.68]).<sup>99</sup> A National Cancer Database study using multivariable-adjusted models to analyze the clinical treatment of 89 319 women with invasive endometrioid endometrial cancer between 2004 and 2014 reported that 75.3% of non-Hispanic White women received NCCN guideline-concordant treatment compared with 70.1% of non-Hispanic Black women (adjusted OR, 0.92 [95% CI, 0.86-0.98]) and 71.0% of Hispanic women (adjusted OR, 0.90 [95% CI, 0.83-0.97]).<sup>100</sup>

### Practical Considerations

Women should follow up with a gynecologic oncologist for at least 2 years after completion of treatment for low-risk endometrial cancer and for at least 5 years after completion of treatment for high-risk disease. Although there is insufficient evidence to support routine use of vaginal cytology, radiographic imaging, or the CA125 blood test as part of surveillance, these tests may be considered if recurrence is suspected.<sup>101</sup> Ongoing primary care should include age- and risk-appropriate screening for other cancers, including breast and colorectal cancers.

A systematic review of 27 observational studies examined patient-reported quality of life outcomes in women with endometrial cancer (individual study size: 38 to 666 patients), with most studies assessing outcomes for less than 2 years.<sup>102</sup> Factors associated with improved quality of life included laparoscopic surgery compared with laparotomy,<sup>102</sup> radiation administered as vaginal brachytherapy rather than external beam pelvic radiation,<sup>102</sup> and sentinel lymph node dissection instead of full pelvic lymphadenectomy.<sup>102,103</sup> Sexual health dysfunction has been reported in approximately three-quarters of patients with endometrial cancer after treatment.<sup>104</sup> Recommended interventions include sexual health discussions, vaginal lubricants and moisturizers, and referral to pelvic floor physical therapy and specialty sexual health clinics.<sup>105,106</sup> NCCN guidelines for survivorship also serve as a resource for clinicians.<sup>107</sup>

### Limitations

This review has several limitations. First, it is not a systematic review and the quality of included literature was not formally evaluated. Second, some relevant studies may have been missed. Third, molecular subtyping and management of endometrial cancer are rapidly evolving, so treatment recommendations may change as the field advances.

### Conclusions

Endometrial cancer accounts for 95% of uterine cancers, which are the fourth most common cancer in women in the US. Surgery is first-line treatment. Adjuvant radiation is recommended for early-stage,

high-intermediate risk endometrial cancer. Adjuvant chemotherapy with carboplatin and paclitaxel is recommended for early-stage disease with aggressive histologic subtypes and advanced-stage disease, which is also treated with immunotherapy in some cases.

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